

CampusNest

Smart Hostel Management System

University of Moratuwa
Faculty of Information Technology

IN1101 – Programming Fundamentals
Academic Year 2024/2025



Team Members & Contributions

Registration No.	Assigned Module	Key Responsibilities
254156	Student Management	Student registration, search, update, delete, and sorting operations
254157	Room Management	Room allocation, room availability tracking, status updates, and room management
254158	Fee Management	Fee recording, payment status updates, fee summaries, and reporting
254159	Visitor Management	Visitor registration, visitor tracking, search functions, and visitor reports
254160	Maintenance Management	Maintenance request handling, status tracking, priority management, and maintenance reports

Problem Context



Manual Record Keeping

All student and room data is tracked on paper, causing delays and human error.



Inconsistent Data

Multiple paper registers lead to conflicting records and lost information.



Slow Response Times

Visitor logs, fee tracking, and maintenance requests take hours to process.

Problem Statement

University hostels lack a centralized, digital system to manage students, rooms, fees, visitors, and maintenance — resulting in operational inefficiencies and poor data reliability.

Key Gaps Identified

✗ No centralized digital record system

✗ Human error in manual registers

✗ No structured fee tracking

✗ No visitor log management

✗ Maintenance requests untracked

Proposed Solution

CampusNest A Console Based C System



C Language

Pure C99

no external libraries, no frameworks. Exactly at PF level.



File Persistence

All data saved to .txt files. Persists between runs. No database needed.



Modular Design

5 independent modules, each with its own .c file and clear responsibility.

Project Objectives

01



Student Management

Register, track, search, and deactivate hostel residents.

02



Room Management

Configure rooms, manage capacity, and track allocations.

03



Fee Management

Record monthly fees, mark payments, view outstanding dues.

04



Visitor Management

Log all visitor entries with contact and purpose details.

05



Maintenance Tracking

Raise, monitor, and resolve hostel maintenance requests.

System Modules



student.c



students.txt

Add / View / Search / Delete



room.c



rooms.txt

Add / View / Allocate / Free



fee.c



fees.txt

Add / View / Pay / Unpaid



visitor.c



visitors.txt

Log / View / Search



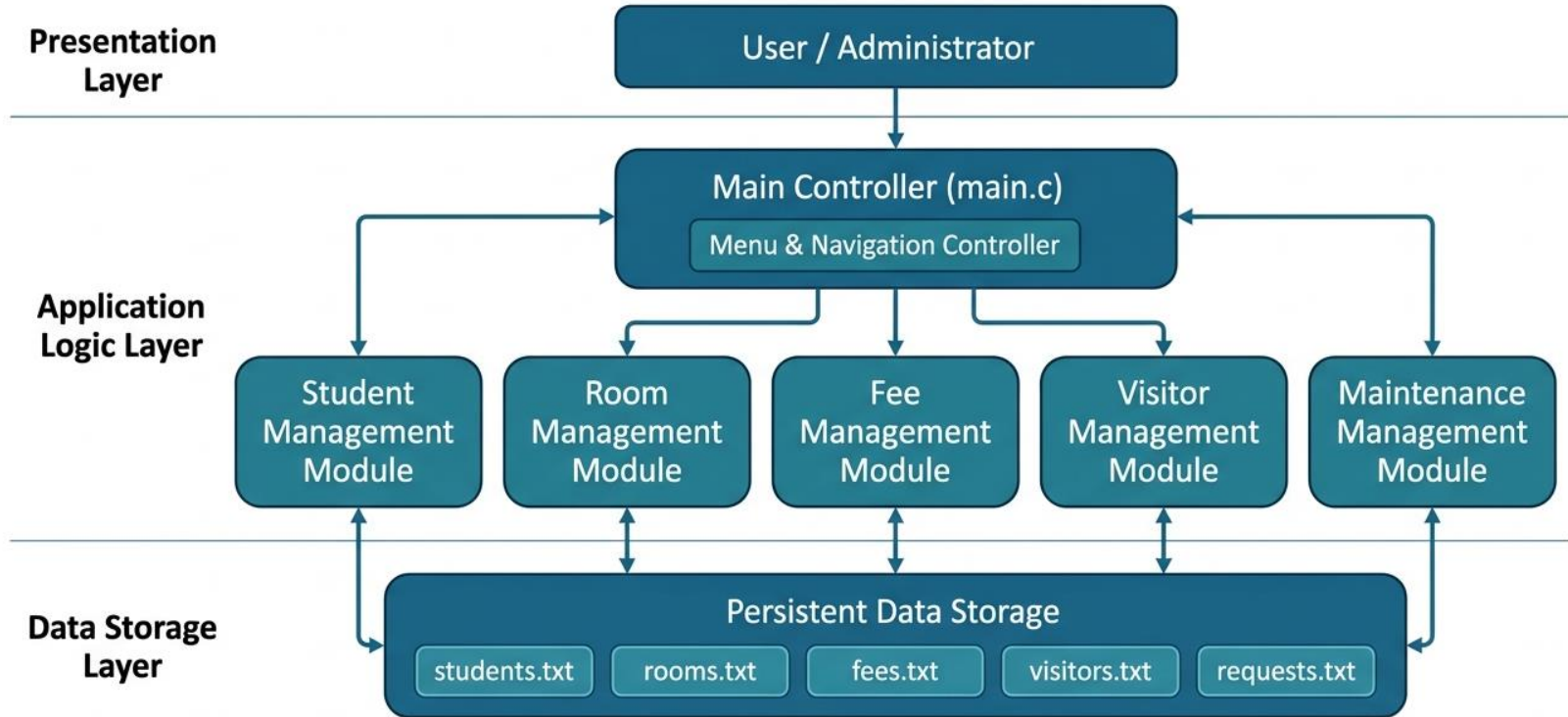
maintenance.c



requests.txt

Log / View / Resolve / Pending

System Architecture



Technical Stack

C Language (C99)

Core programming language. Structs, arrays, pointers, and standard I/O.

Structures

typedef struct — Student, Room, FeeRecord, Visitor, MaintenanceRequest.

Control Flow

while loop + switch-case for menu-driven interaction and safety.

File Handling

fopen / fclose / fprintf / fscanf — append and read for persistence.

Functions

Modular decomposition with single-responsibility per function.

Input Validation

fgets + sscanf pattern — prevents buffer overflow and bad input.

System Execution Evidence

```
--- CAMPUSNEST: MAIN MENU ---
```

- 1) Student Management
- 2) Room Management
- 3) Fee Management
- 4) Visitor Management
- 5) Maintenance Mgmt
- 0) Exit

```
-----
```

```
Choice: 1
```

Figure 1 – CampusNest Main Menu Interface

```
=== MEMBER 1: STUDENT MANAGEMENT ===
```

- 1) Add Student
- 2) View All Students
- 3) Search Student
- 4) Update Room Assignment
- 5) Delete Student
- 6) Sort by ID
- 0) Back to Main Menu

```
-----  
Choice: 1
```

```
Student ID      : 254156  
Full Name       : iman perera  
Age             : 20  
Gender (M/F)    : M  
Contact No      : 0717207318  
Email           : imanperera@gmail.com  
Room Number     : 11  
Enroll Date     : 2026/04/11  
[+] Student 'iman perera' added successfully.
```

```
=== MEMBER 1: STUDENT MANAGEMENT ===
```

- 1) Add Student
- 2) View All Students
- 3) Search Student
- 4) Update Room Assignment
- 5) Delete Student
- 6) Sort by ID
- 0) Back to Main Menu

```
-----  
Choice: 1
```

```
Student ID      : 254157  
Full Name       : savindu perera  
Age             : 20  
Gender (M/F)    : M  
Contact No      : 0705921490  
Email           : savinduperera@gmail.com  
Room Number     : 12  
Enroll Date     : 2024/06/08  
[+] Student 'savindu perera' added successfully.
```

Figure 2 – Student Records Display and add Student Input Interface

=== MEMBER 1: STUDENT MANAGEMENT ===

- 1) Add Student
- 2) View All Students
- 3) Search Student
- 4) Update Room Assignment
- 5) Delete Student
- 6) Sort by ID
- 0) Back to Main Menu

Choice: 3

Search by: 1) ID 2) Name

Choice: 1

Enter ID: 254157

ID : 254157
Name : savindu perera
Age : 20
Gender : M
Contact : 0705921490
Email : savinduperera@gmail.com
Room : 12
Enrolled: 2024/06/08

=== MEMBER 1: STUDENT MANAGEMENT ===

- 1) Add Student
- 2) View All Students
- 3) Search Student
- 4) Update Room Assignment
- 5) Delete Student
- 6) Sort by ID
- 0) Back to Main Menu

Choice: 3

Search by: 1) ID 2) Name

Choice: 2

Enter Name: savindu perera

ID	Name	Age	Room

254157	savindu perera	20	12

Figure 4 – Student Search Result

```
=== MEMBER 1: STUDENT MANAGEMENT ===
```

- 1) Add Student
- 2) View All Students
- 3) Search Student
- 4) Update Room Assignment
- 5) Delete Student
- 6) Sort by ID
- 0) Back to Main Menu

```
-----  
Choice: 5
```

```
Enter Student ID to delete: 254157
```

```
[+] Student deleted.
```

```
=== MEMBER 1: STUDENT MANAGEMENT ===
```

- 1) Add Student
- 2) View All Students
- 3) Search Student
- 4) Update Room Assignment
- 5) Delete Student
- 6) Sort by ID
- 0) Back to Main Menu

```
-----  
Choice: 3
```

```
Search by: 1) ID    2) Name
```

```
Choice: 1
```

```
Enter ID: 254157
```

```
No matching student found.
```

Figure 5 – Student Deletion Confirmation

```
--- CAMPUSNEST: MAIN MENU ---
```

- 1) Student Management
- 2) Room Management
- 3) Fee Management
- 4) Visitor Management
- 5) Maintenance Mgmt
- 0) Exit

```
-----  
Choice: 2
```

```
=== MEMBER 2: ROOM MANAGEMENT ===
```

- 1) Add Room
- 2) View All Rooms
- 3) Search Room
- 4) Update Room Status
- 5) View Available Rooms
- 6) Sort Rooms
- 7) Delete Room
- 0) Back to Main Menu

```
-----  
Choice: █
```

Figure 6 – Room Records Display

```
=== MEMBER 1: STUDENT MANAGEMENT ===
```

- 1) Add Student
- 2) View All Students
- 3) Search Student
- 4) Update Room Assignment
- 5) Delete Student
- 6) Sort by ID
- 0) Back to Main Menu

```
-----  
Choice: 5
```

```
Enter Student ID to delete: 254157
```

```
[+] Student deleted.
```

```
=== MEMBER 1: STUDENT MANAGEMENT ===
```

- 1) Add Student
- 2) View All Students
- 3) Search Student
- 4) Update Room Assignment
- 5) Delete Student
- 6) Sort by ID
- 0) Back to Main Menu

```
-----  
Choice: 3
```

```
Search by: 1) ID    2) Name
```

```
Choice: 1
```

```
Enter ID: 254157
```

```
No matching student found.
```

Figure 5 – Student Deletion Confirmation

```
--- CAMPUSNEST: MAIN MENU ---
```

- 1) Student Management
- 2) Room Management
- 3) Fee Management
- 4) Visitor Management
- 5) Maintenance Mgmt
- 0) Exit

```
-----  
Choice: 2
```

```
=== MEMBER 2: ROOM MANAGEMENT ===
```

- 1) Add Room
- 2) View All Rooms
- 3) Search Room
- 4) Update Room Status
- 5) View Available Rooms
- 6) Sort Rooms
- 7) Delete Room
- 0) Back to Main Menu

```
-----  
Choice: █
```

Figure 6 – Room Records Display

```
=== MEMBER 2: ROOM MANAGEMENT ===
```

- 1) Add Room
- 2) View All Rooms
- 3) Search Room
- 4) Update Room Status
- 5) View Available Rooms
- 6) Sort Rooms
- 7) Delete Room
- 0) Back to Main Menu

```
-----  
Choice: 1
```

```
Room Number   : 18
```

```
Type (Single/Double/Triple): Single
```

```
Capacity      : 2
```

```
Price/Month   : 5000
```

```
[+] Room 18 added.
```

```
=== MEMBER 2: ROOM MANAGEMENT ===
```

- 1) Add Room
- 2) View All Rooms
- 3) Search Room
- 4) Update Room Status
- 5) View Available Rooms
- 6) Sort Rooms
- 7) Delete Room
- 0) Back to Main Menu

```
-----  
Choice: 1
```

```
Room Number   : 29
```

```
Type (Single/Double/Triple): Double
```

```
Capacity      : 4
```

```
Price/Month   : 10000
```

```
[+] Room 29 added.
```

```
=== MEMBER 2: ROOM MANAGEMENT ===
```

- 1) Add Room
- 2) View All Rooms
- 3) Search Room
- 4) Update Room Status
- 5) View Available Rooms
- 6) Sort Rooms
- 7) Delete Room
- 0) Back to Main Menu

```
-----  
Choice: 2
```

Room	Type	Capacity	Occupied	Status	Price/Month
18	Single	2	0	Available	5000.00
29	Double	4	0	Available	10000.00

```
-----  
Total rooms: 2
```

Figure 7 – Add Room Interface

```
--- CAMPUSNEST: MAIN MENU ---
1) Student Management
2) Room Management
3) Fee Management
4) Visitor Management
5) Maintenance Mgmt
0) Exit
-----
Choice: 3

=== MEMBER 3: FEE MANAGEMENT ===
1) Add Fee Record
2) View All Fees
3) Search by Student ID
4) Update Fee Status
5) Fee Summary
6) Sort by Amount
0) Back to Main Menu
-----
Choice: █
```

Figure 8 – Fee Records Display

```
--- CAMPUSNEST: MAIN MENU ---
1) Student Management
2) Room Management
3) Fee Management
4) Visitor Management
5) Maintenance Mgmt
0) Exit
-----
Choice: 4

=== MEMBER 4: VISITOR MANAGEMENT ===
1) Log Visitor
2) View All Visitors
3) Search by Student ID
4) Visitor Count Report
5) Sort by Date
6) Delete Visitor Record
0) Back to Main Menu
-----
Choice: █
```

Figure 9 – Visitor Records Display

```
--- CAMPUSNEST: MAIN MENU ---
1) Student Management
2) Room Management
3) Fee Management
4) Visitor Management
5) Maintenance Mgmt
0) Exit
-----
Choice: 5

=== MEMBER 5: MAINTENANCE MANAGEMENT ===
1) Add Request
2) View All Requests
3) Update Request Status
4) Search by Room
5) Maintenance Summary
6) Sort by Priority
7) Delete Request
0) Back to Main Menu
-----
Choice: █
```

Figure 11 – Maintenance Requests Display

System Benefits



Centralised Records

All hostel data in one system — students, rooms, fees, visitors, maintenance.



Faster Operations

Instant search, allocation, and report generation via menu-driven interface.



Reduced Human Error

Structured input and validation eliminates data inconsistency.



Data Persistence

Text-file storage ensures data survives across all program restarts.



Audit Trail

All fee payments, visitor logs, and maintenance requests are permanently recorded.



Easy Maintainability

Modular .c files allow updates to one module without affecting others.

Conclusion

CampusNest is a console-based hostel management system developed to simplify the management of students, rooms, fees, visitors, and maintenance requests. The project successfully applies key C programming concepts such as structures, functions, arrays, searching, sorting, and input validation to solve a real-world problem.

Thank You!

CampusNest – Smart Hostel Management System

Questions?